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=====
                        DARKCAT – FIELD INSTRUCTIONS
Concrete examples for every supported network
and every major feature
=====
```

This file is a tour of darkcat by example. Every command shown here is real and runnable; every URL is from darkcat's built-in seeds (see `darkcat list all`). Output snippets are abridged and trimmed for readability.

Conventions:

```
$ denotes a shell prompt
> denotes the darkcat REPL prompt
... abridges long output
Lines starting with `#` are comments inside an example.
```

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```
=====
1. INSTALL AND FIRST-RUN HEALTH CHECK
=====
```

```
$ python -m venv .venv && . .venv/bin/activate
$ pip install -e .

# Bootstrap config / dirs / seed lists.
$ darkcat init

# Any time after install, sanity-check the environment.
$ darkcat doctor
```

Expected output (abridged):

```
● darkcat doctor – environment health
[ok] home directory      /home/you/.darkcat
[ok] sqlite database     ~/.darkcat/crawl.db (rwx)
[ok] tor SOCKS           127.0.0.1:9050 reachable
[warn] i2p HTTP proxy    127.0.0.1:4444 NOT reachable
[ok] ipfs gateway        127.0.0.1:8080 reachable
[warn] yggdrasil         no 200::/7 route
[ok] gnupg               gpg 2.4.4
[ok] beautifulsoup4 / lxml / rich / textual
[ok] chat backends       6/9 ready (missing: tox briar ricochet)
[ok] proton bridge       127.0.0.1:1025 listening (or [warn] refused)
[ok] mail hosts          smtp disroot.org:587 reachable
[ok] mail hosts          imap disroot.org:993 reachable
```

If a daemon is missing darkcat keeps working – it just won't crawl that

protocol. Install only what you actually need.

The three summary lines at the bottom are deliberately small. The chat-backends row aggregates `darkcat chat backends` into one entry so the doctor table doesn't bloat by one row per backend. The proton-bridge row is a 400ms TCP probe of 127.0.0.1:1025 – [ok] only when a local Proton Mail Bridge is actually listening (used by `mail send / check` against ProtonMail personas). The mail-hosts rows walk every persona in the vault, collect each SMTP coordinate (`site`) + IMAP coordinate (parsed from `notes`), and run a 2 s `socket.create\_connection` against each – surfaces dead presets before `mail send` blows up with an opaque socket error. Encrypted vaults are skipped here (decrypt first with `darkcat personas decrypt` if you want this check).

2. DISCOVER WHAT DARKCAT CAN REACH

```
# Quick, machine-friendly status.
$ darkcat status

  protocol  reachable  notes
  tor       yes     127.0.0.1:9050
  i2p       no      start i2pd or i2p
  ipfs      yes     kubo gateway
  freenet   no      start FProxy on 8888
  gemini    yes     direct (TLS+TOFU)
  gopher    yes     direct
  clearnet  yes     (tunnelled through tor)

# Print every curated entry point we ship – handy as a starter map.
$ darkcat list all | head -40
$ darkcat list tor          # only tor entries
$ darkcat -la               # short alias for list-all

# Print the raw seed URLs (no name / desc columns).
$ darkcat seeds tor
$ darkcat seeds i2p
$ darkcat seeds all > /tmp/seeds.txt
```

3. TOR – FINDING HACKING, CRYPTO AND FORUM CONTENT

Goal: find darknet forums on hacking and cryptocurrency, crawl what looks relevant, then search across the result.

```
# Step 1 – discover seed URLs by topic. `discover` queries Ahmia,
# tor.taxi, dark.fail, onion.live and similar directories through tor
# and aggregates onion URLs that match. The query is positional.
$ darkcat discover "hacking forum"
$ darkcat discover "cryptocurrency mixer"
$ darkcat discover "carding forum" --max-per-engine 30

# Result lines look like:
# [tor.taxi] http://abc...xyz.onion/ "DarkForum – eng/rus hacking"
# [ahmia]    http://def...uvw.onion/ "Crypto mixer review board"
#
# Save the URLs to a file for the next step (one URL per line, ready
# to feed straight into --seed-file).
$ darkcat discover "hacking forum" --max-per-engine 50 \
  > /tmp/hacking_seeds.txt

# Step 2 – crawl those seeds, BFS depth-2, with topic scoring so pages
# mentioning your keywords score higher.
$ darkcat crawl --seed-file /tmp/hacking_seeds.txt \
  --max-depth 2 --max-pages 400 \
  --topics exploit cve 0day payload phishing rat leak

# Step 3 – search what you crawled, FTS5.
$ darkcat search "0day AND windows"
$ darkcat search "monero mixer"
$ darkcat search "wallet drainer NEAR phishing"      # FTS5 NEAR
```

```
$ darkcat search "carding" --strict # AND of phrases
```

```
# Step 4 – show the highest-scoring pages from the crawl.
```

```
$ darkcat top --limit 25
```

```
$ darkcat top --protocol tor --limit 50
```

A few well-documented public seeds to test the pipeline (these are stable clearnet directories that themselves index onions):

```
https://tor.taxi/      tor onion directory
https://dark.fail/    curated onion uptime monitor
https://ahmia.fi/     clearnet onion search engine
https://thehiddenwiki.com/ legacy onion index
http://duckduckgogg42xjoc72x3sjasowarfbgcmvfimaftt6twagswzczad.onion/
                        DuckDuckGo's own onion service
```

```
# Crawl just the big four directories to seed your local index.
```

```
$ darkcat crawl --seeds https://tor.taxi/ https://dark.fail/ \
                https://ahmia.fi/ \
                --max-depth 1 --max-pages 200
```

4. I2P – EEPSITES AND RESEED

I2P needs a local I2P router. On Fedora:

```
$ sudo dnf install i2pd
$ sudo systemctl enable --now i2pd
$ darkcat status | grep i2p # should now say 'yes'
```

```
# I2P seeds shipped with darkcat:
```

```
# notbob.i2p      uptime + index
# identiguy.i2p   eepsite directory
# reg.i2p         domain registry
# stats.i2p       router statistics
# legwork.i2p     eepsite search
```

```
# Crawl the directory pages to populate your local index.
```

```
$ darkcat crawl --seeds http://notbob.i2p/ http://identiguy.i2p/ \
                --max-depth 2 --max-pages 300 \
                --topics i2p eepsite onion privacy
```

```
# Single-shot fetch (one URL, transport auto-picked from suffix):
```

```
$ darkcat fetch http://notbob.i2p/
```

```
# I2P jump-service auto-resolve: when a fresh .i2p hostname doesn't
```

```
# have a local subscription yet, darkcat falls back to a jump service
```

```
# (stats.i2p) so the fetch still resolves on first hit.
```

```
$ darkcat fetch http://example-new.i2p/
```

```
# Search across what you crawled (FTS5 across all protocols):
```

```
$ darkcat search "anonymity"
```

```
$ darkcat search "anonymity" --strict # AND of phrases
```

5. IPFS / IPNS – CONTENT-ADDRESSED WEB

Needs a local Kubo node (ipfs daemon). The default gateway is 127.0.0.1:8080. If you don't run one, darkcat falls back to a public gateway (configurable in `~/.darkcat/config.toml`).

```
# Real IPNS names you can crawl right now:
```

```
$ darkcat fetch ipns://docs.ipfs.tech
```

```
$ darkcat fetch ipns://en.wikipedia-on-ipfs.org
```

```
$ darkcat fetch ipns://blog.ipfs.tech
```

```
# Direct CID:
```

```
$ darkcat fetch ipfs://bafybeibwzifw52ttrkqlikfzext5akxu7lz4xiwjgwmqcpdzmp3n5vnbe
```

```
# Walk a small bundle: BFS depth-2 from the IPFS docs root.
```

```
$ darkcat crawl --seeds ipns://docs.ipfs.tech \
```

```
--max-depth 2 --max-pages 150
```

```
# Read-only mirror of Wikipedia on IPFS – perfect for offline research:
```

```
$ darkcat crawl --seeds ipns://en.wikipedia-on-ipfs.org \  
--max-depth 1 --max-pages 50 \  
--topics encryption privacy opsec
```

```
# Search the result.
```

```
$ darkcat search "end-to-end encryption"
```

6. FREENET / HYPHANET – LONG-FORM ANONYMOUS PUBLISHING

Needs FProxy on 127.0.0.1:8888 (the Hyphanet node web UI port).

```
# Two real Hyphanet indices (USK = Updateable Subspace Key):
```

```
$ darkcat fetch \  
"freenet:USK@XJZAi25dd5y7lrxE3cHmM-xZ-c-hlPpKLYeLC0YG5I,8XTbR1bd9RBXLX6j-0ZNednsJ8Cl6E  
AeBBebC3jtMFU,AQACAAE/index/-1/"
```

```
# That's "Enzo's Index" – a manually curated freesite directory.
```

```
$ darkcat fetch \  
"freenet:USK@7H7sH5lEeFD80sigMb35GaCD03N50CUKgi9D0iuxn7E,3SBfWf-DvMCwFmIfVx3lxYDEJpRobr  
cYZ4DbtUq3MbA,AQACAAE/linkageddon/-1/"
```

```
# "Linkageddon" – auto-generated freesite firehose.
```

```
# Crawl the indices to bootstrap a pile of freesites.
```

```
$ darkcat crawl --seeds \  
"freenet:USK@XJZAi25dd5y7lrxE3cHmM-xZ-c-hlPpKLYeLC0YG5I,8XTbR1bd9RBXLX6j-0ZNednsJ8Cl6E  
AeBBebC3jtMFU,AQACAAE/index/-1/" \  
--max-depth 1 --max-pages 60
```

Freenet keys come in CHK (immutable hash), SSK (signed mutable),
USK (versioned signed), KSK (legacy keyword) flavours; darkcat treats
all four uniformly.

7. ZERONET – PEER-TO-PEER MIRRORS

Needs a ZeroNet UI on 127.0.0.1:43110.

```
# The canonical "ZeroHello" entry page:
```

```
$ darkcat fetch zero://1HelloAddr8MZUPYMTj9CSJj3wfGRjEHr
```

```
# Walk content.json for one site to discover its files.
```

```
$ darkcat zeronet-walk 1HelloAddr8MZUPYMTj9CSJj3wfGRjEHr
```

8. GEMINI AND GOPHER – THE SMALL WEB

No daemon needed. Gemini speaks TLS+TOFU directly; Gopher speaks raw TCP.

```
# Real Gemini capsules:
```

```
$ darkcat fetch gemini://geminiprotocol.net/  
$ darkcat fetch gemini://geminispace.info/      # search engine  
$ darkcat fetch gemini://kennedy.gemi.dev/      # crawler / search
```

```
# Crawl Geminispace from a single seed.
```

```
$ darkcat crawl --seeds gemini://geminispace.info/ \  
--max-depth 2 --max-pages 200 \  
--topics phlog smolnet privacy
```

```
# Real Gopher holes:
```

```
$ darkcat fetch gopher://gopher.floodgap.com/  
$ darkcat fetch gopher://sdf.org/  
$ darkcat fetch gopher://gopher.club/
```

```
# Crawl a multi-host pool.
```

```
$ darkcat crawl --seeds gopher://gopher.floodgap.com/ \  
gopher://sdf.org/ \  
gopher://gopher.club/
```

```
gopher://sdf.org/ gopher://gopher.club/ \
--max-depth 2 --max-pages 100
```

9. ENS, HANDSHAKE, OPENNIC, NAMECOIN – ALT-ROOTS

```
# ENS .eth via eth.limo gateway (no local resolver needed):
$ darkcat fetch https://vitalik.eth/
$ darkcat fetch https://nick.eth/

# Handshake (HNS) – needs hsd/hnsd locally, else hns.is gateway:
$ darkcat fetch http://welcome.nb/
$ darkcat fetch http://proofofconcept/

# OpenNIC peering TLDs – needs an OpenNIC DNS server in /etc/resolv.conf
# or via systemd-resolved drop-in:
$ darkcat fetch http://opennic.geek/
$ darkcat fetch http://wiki.opennic.geek/
$ darkcat fetch http://grep.geek/          # OpenNIC search

# Namecoin (.bit) – needs ncdns:
$ darkcat fetch http://nx.bit/

# Alt-root bulk crawl:
$ darkcat crawl --seeds https://vitalik.eth/ http://opennic.geek/ \
    http://wiki.opennic.geek/ http://welcome.nb/ \
    --max-depth 1 --max-pages 80
```

10. SEARCH AND ANALYSE WHAT YOU HAVE CRAWLED

Everything goes into ~/.darkcat/crawl.db (SQLite + FTS5).

```
# Full-text search.
$ darkcat search "ransomware leak"
$ darkcat search "monero AND mixer"
$ darkcat search "phishing OR drainer"

# Strict mode (AND of phrases instead of broad prefix-OR matching).
$ darkcat search "wallet drainer" --strict

# Cap result count.
$ darkcat search "0day" --limit 100

# Top-scoring pages overall (uses --topics scores recorded at crawl time).
$ darkcat top --limit 30
$ darkcat top --protocol i2p

# Database statistics.
$ darkcat stats

    pages          12 348
   findings         4 091
  protocols      tor=8211 i2p=423 ipfs=1106 freenet=88 ...
   topics          0day(412) ransomware(391) monero(284) ...

# See snapshots over time of one URL.
$ darkcat history --url "https://tor.taxi/"

# Show diffs across crawl windows for that URL.
$ darkcat diff --url "https://tor.taxi/"
```

11. WATCHLIST AND ALERTS

A watchlist entry fires an alert when a future crawl/scan matches it. Match fields are `--target` (substring on the finding's target), `--category` (category name), and `--sample` (substring on the raw sample). Add `--regex` to treat any of those as regex. Every rule needs a `--sink`.

```
# 1) Sample-text keyword "0day" → log line on every match.
$ darkcat watch add --sample 0day --sink log

# 2) Regex on raw card numbers → desktop notification.
$ darkcat watch add --sample '\b\d{16}\b' --regex --sink notify

# 3) Target-domain match on tor.taxi → append JSON to file.
$ darkcat watch add --target tor.taxi --sink file:/tmp/alerts.jsonl

# List current rules.
$ darkcat watch list
```

id	target	category	sample	regex	sink
1	-	-	0day	no	log
2	-	-	\b\d{16}\b	yes	notify
3	tor.taxi	-	-	no	file:/tmp/alerts.jsonl

```
# Test sink #1 by synthesising a finding (use the id from `watch list`).
$ darkcat watch test 1

# Read alerts that fired during recent crawls.
$ darkcat alerts --limit 100
```

12. DIFF: CHANGE-WATCHING ONE URL OVER TIME

```
# Snapshot 1.
$ darkcat fetch https://tor.taxi/

# ... later ...
$ darkcat fetch https://tor.taxi/

# Show what changed between snapshots.
$ darkcat diff --url "https://tor.taxi/"

@@ +5 lines added @@
+ http://newonion12345.onion/    "Just listed"
@@ -2 lines removed @@
- http://gone67890.onion/      "Down"
```

Combine with `darkcat schedule` for autonomous monitoring.

13. PGP KEY HARVEST

`darkcat keys harvest` walks crawled pages, extracts PGP public-key blocks, and indexes them in the DB. Useful for OSINT (link a handle to a key, watch for re-use across sites).

```
$ darkcat keys harvest          # extract from existing crawl
$ darkcat keys list
$ darkcat keys list --fpr DEADBEEF    # filter by fingerprint substring
$ darkcat keys show 0xDEADBEEFCAFE1234 # one key by long-id
```

keyid	0xDEADBEEFCAFE1234
uids	"Alice <alice@example.onion>"
seen at	http://forum.onion/u/alice/profile
	http://leaks.onion/dump/2026-04
first	2026-04-12
last	2026-05-03

14. MIRROR CLUSTERING (FIND CLONES / PHISHES)

Same content reachable via several URLs (legit mirrors *or* phishes that clone the page). `darkcat clusters` groups pages by content fingerprint.

```
$ darkcat clusters --min 3 --limit 20
```

```
cluster #1 (4 pages, fp=ab12cd34...)
http://realmarket.onion/
http://realmarket-mirror1.onion/
http://realmarket-mirror2.onion/
http://phishclone789.onion/ ← suspicious, registered 2026-04-29
```

```
$ darkcat clusters --json --limit 50 > /tmp/mirrors.json
```

15. IMAGE OCR

Some forums / paste sites post scans / screenshots; OCR pulls text out.
`darkcat ocr` takes a URL – it fetches the page, runs OCR over inline images, and prints the extracted text.

```
# OCR every image embedded on a page.
$ darkcat ocr https://tor.taxi/
```

```
# Cap how many images we OCR per page (default unlimited):
$ darkcat ocr --max-images 5 https://tor.taxi/
```

```
# Override Tesseract language pack (default eng).
$ darkcat ocr --lang spa https://example.onion/
```

OCR backend is Tesseract; install `tesseract` from your distro.

16. TOR CIRCUITS (NEWNYM, INFO, BRIDGES)

Needs Tor's ControlPort enabled (CookieAuth or HashedControlPassword).

```
# Force a fresh circuit.
$ darkcat tor newnym
```

```
# Show current circuits.
$ darkcat tor circuits
```

id	path	state
212	guard1 → middle3 → exit7	BUILT
213	guard1 → middle9 → exit2	EXTENDED

```
# Lookup an onion descriptor.
$ darkcat tor descriptor duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion
```

```
# Bridge management.
$ darkcat tor bridges
$ darkcat tor bridges-add "obfs4 1.2.3.4:443 cert=..."
$ darkcat tor bridges-clear
```

17. LIVENESS PROBING AND LOOP MONITORING

Periodically check that previously-crawled URLs are still up.

```
# One pass over a list of URLs.
$ darkcat liveness probe \
  http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/ \
  https://tor.taxi/
```

```
● [tor      ] 200 480ms 18420B http://...onion/
● [clearnet ] 200 110ms  9821B https://tor.taxi/
```

```
# Probe everything we know about.
$ darkcat liveness probe --known --limit 200
```

```
# Run a continuous loop, every 30 min:
$ darkcat liveness loop --known --interval 1800
```

```
# Restrict to one protocol:
$ darkcat liveness probe --known --protocol i2p --limit 50
```

18. SCHEDULED RE-CRAWLS

```
# Add a recurring crawl every 6 hours. `name` is positional;
# --interval is in seconds; --seeds takes URLs.
$ darkcat schedule add hacking-forums \
    --interval 21600 \
    --seeds https://tor.taxi/ https://dark.fail/ \
    https://ahmia.fi/ \
    --max-depth 2 --max-pages 200 \
    --topics exploit cve 0day payload

$ darkcat schedule list
$ darkcat schedule run-due      # one-shot kick of due jobs
$ darkcat schedule loop --tick 60 # daemon-mode runner
$ darkcat schedule disable hacking-forums
$ darkcat schedule remove hacking-forums
```

19. EXPORT AND IOC FORMATS (JSONL, STIX 2.1, MISP)

```
# Export findings as JSONL, last 7 days.
$ darkcat export --format jsonl --since 7d > /tmp/recent.jsonl

# Export findings as STIX 2.1.
$ darkcat export --format stix > /tmp/iocs.stix.json

# Export findings as MISP-compatible events.
$ darkcat export --format misp > /tmp/iocs.misp.json

# Useful pivot – extract every onion seen in the last 30 days, dedupe.
$ darkcat export --format jsonl --since 30d \
    | jq -r 'select(.protocol=="tor") | .url' | sort -u
```

20. CHAT NETWORKS (TELEGRAM, MATRIX, XMPP, ...)

Each chat backend is auto-detected; `darkcat chat backends` reports which dependencies are present.

```
$ darkcat chat backends
```

network	ready	dependency
telegram	yes	telethon
matrix	yes	matrix-nio
xmpp	no	slixmpp (pip install slixmpp)
simplex	no	simplex-cli
session	no	session-cli
tox	no	pytox

Step-by-step: log into Telegram with a darkcat persona, list channels, read one, ingest into the search index.

```
# 1. Create a persona dedicated to a given Telegram account.
# Persona name is positional.
$ darkcat personas add tg-research \
    --network telegram \
    --site api.telegram.org \
    --handle "+34xxxxxxxxxx"

# 2. Login. First call prompts for the SMS code interactively.
$ darkcat chat login --persona tg-research --network telegram

# 3. List channels.
$ darkcat chat list --persona tg-research --limit 30
```


id	kind	name	people	unread
-1001234..	channel	Crypto News (RU)	4231	7
-1009876..	group	OSINT EU	218	3

4. Read recent messages. channel_id is positional.

```
$ darkcat chat read --persona tg-research \
    --limit 50 -- -1001234567890
```

5. Ingest the channel into ~/.darkcat/crawl.db so `search` finds it.

```
$ darkcat chat ingest --persona tg-research \
    --limit 500 -- -1001234567890
$ darkcat search "exchange listing"
```

TUI and GUI chat consoles ship a row of preset buttons so you don't have to remember which CLI verb each network uses:

Telegram Join	Telegram Leave	Add Session	Accept SimpleX
Login	List	Backends	

Each button just pre-fills the Network + Action dropdowns (and shows a one-line hint for the expected target syntax) – press Run to execute.

The five chat networks whose clients darkcat does NOT bundle still get identity-vault entries so you can rotate / burn them like every other account:

matrix	federated; multiple homeservers – pick with `--instance`
xmpp	federated; multiple servers – pick with `--instance`
tox	local keypair (qTox) – no central server, no recovery
briar	local keypair (Briar app) – peer-to-peer, briar:// links
ricochet	Tor v3 onion identity (Ricochet Refresh) – desktop GUI only

`darkcat chat backends` will list these as "no client bundled" while the identity vault still tracks the account. Conversation itself happens in the official desktop apps; darkcat handles signup hygiene + linking.

21. IDENTITY GENERATOR – FULL TOUR

Compartmented account creation. One persona per project. Manual-assist signup through Tor / I2P / proxy / VPN-pin. Soft cap of 5 active personas per provider (cap doesn't count `burned`).

21.1 Available providers

```
$ darkcat identity providers
```

slug	name	category	transport
protonmail	ProtonMail	email	tor
tutanota	Tutanota	email	tor
mailfence	Mailfence	email	tor
disroot	Disroot	email	tor
gmail	Gmail	webmail	vpn-pin
outlook	Outlook	webmail	vpn-pin
yahoo	Yahoo	webmail	vpn-pin
protonvpn	ProtonVPN	vpn	tor
tunnelbear	TunnelBear	vpn	tor
windscribe	Windscribe	vpn	tor
reddit	Reddit	social	tor
twitter	X / Twitter	social	vpn-pin
telegram	Telegram	social	vpn-pin
discord	Discord	social	vpn-pin
mastodon	Mastodon	social	tor
matrix	Matrix	social	tor (multi-instance)
xmpp	XMPP / Jabber	social	tor (multi-instance)
session	Session	social	tor
simplex	SimpleX	social	tor
tox	Tox	social	tor (local keygen)
briar	Briar	social	tor (local keygen)
ricochet	Ricochet Refresh	social	tor (onion keygen)

Drill into one profile (signup URL, fields, no-phone path, ToS).

```
$ darkcat identity providers --slug protonmail
$ darkcat identity providers --slug matrix      # see all homeservers
$ darkcat identity providers --slug xmpp        # see all servers
```

21.2 Create a ProtonMail identity

```
$ darkcat identity new --provider protonmail \
    --transport tor \
    --purpose "research-forum-X"

+ identity protonmail-quiet_river_4821 created (protonmail/email)
  handle      quiet_river_4821
  password    8x2-fJW9_zT...vGqM (shown once – store now)
  display_name Mira Holt
  locale/tz   en_US / America/New_York
  birthdate   1991-07-14
  bio         security & privacy • EU-based, mostly evenings – no DMs
  transport   tor (auth=4f2a91e1c8b3d7a0)
  signup_url  https://account.proton.me/signup

No-phone path: use the onion (protonmailrmez3lotcc...onion)
and select email-only verification.
Status: pending – run `darkcat identity confirm <name>` after signup.

# Open the signup page through Tor (Tor Browser if installed, else
# your default browser with HTTPS_PROXY=socks5h://127.0.0.1:9050):
$ darkcat identity launch protonmail-quiet_river_4821

# Or fold the post-signup capture into one command – `--capture`
# drops you into an inline prompt for the final handle, recovery
# email, and recovery codes the provider just showed once:
$ darkcat identity launch protonmail-quiet_river_4821 --capture

# In the TUI and GUI, pressing Launch / clicking 'Launch signup'
# opens the signup page and, after dismissing the launch block,
# pops the Edit form pre-loaded with the persona – same effect,
# without typing into a terminal.

# After completing signup in the browser:
$ darkcat identity confirm protonmail-quiet_river_4821

# Months later, when the project ends:
$ darkcat identity burn protonmail-quiet_river_4821 \
    --note "project closed 2027-02"
```

Same flow from the GUI: launch `darkcat gui`, menu Identity → Open vault... (Ctrl+Shift+I), click `New`, pick `protonmail`, click `Create`. Select the row, click `Launch signup`. After signup, click `Confirm`.

21.3 Other providers – concrete commands

```
# Tutanota (no phone ever):
$ darkcat identity new --provider tutanota --purpose "newsletter-account"

# Disroot (manual approval – wait 24-48h before confirm):
$ darkcat identity new --provider disroot --purpose "activist-mail"

# Mastodon, with a specific instance:
$ darkcat identity new --provider mastodon --instance fosstodon \
    --purpose "FOSS announcements"
$ darkcat identity new --provider mastodon --instance infosec-exchange \
    --purpose "infosec posting"

# Matrix – pick a homeserver. The signup_url and persona's site/
# network_or_domain are derived from the chosen instance:
$ darkcat identity new --provider matrix --instance tchncs \
    --purpose "research-room-X"
$ darkcat identity new --provider matrix --instance disroot \
    --purpose "activist-room"

# XMPP – pick a server:
$ darkcat identity new --provider xmpp --instance disroot \
    --purpose "private XMPP"
```

```
$ darkcat identity new --provider xmpp --instance jabber-at

# Local-keygen networks – no signup form, the client generates the
# ID; record it in the vault for hygiene + linking, then paste the
# actual ID back via `identity edit` once the client gives it to you:
$ darkcat identity new --provider session --purpose "anon DM"
$ darkcat identity new --provider simplex --purpose "anon DM"
$ darkcat identity new --provider tox --purpose "p2p chat"
$ darkcat identity new --provider briar --purpose "OOB chat"
$ darkcat identity new --provider ricochet --purpose "onion DM"

# Reddit through Tor:
$ darkcat identity new --provider reddit --transport tor \
  --purpose "lurk on r/netsec"

# ProtonVPN (free tier – email only, no phone):
$ darkcat identity new --provider protonvpn --purpose "vpn-egress-EU"
```

If you pass an `--instance` slug the provider does not advertise, the command fails fast with the list of available hosts – no silent fallback to the default homeserver.

21.4 Listing, showing, rotating, linking

```
-----

# List all identities, filter by provider/status/purpose.
$ darkcat identity list
$ darkcat identity list --provider protonmail
$ darkcat identity list --status confirmed
$ darkcat identity list --purpose "research"

# Show one (secrets masked by default).
$ darkcat identity show protonmail-quiet_river_4821
# Reveal secrets – prompts y/N unless DARKCAT_VAULT_PASSPHRASE is set.
$ darkcat identity show protonmail-quiet_river_4821 --reveal

# Rotate the password (old value overwritten).
$ darkcat identity rotate-password protonmail-quiet_river_4821 \
  --print

# Link a recovery account: child = the protected one, parent = recovery.
$ darkcat identity link tutanota-stale_owl_3345 \
  protonmail-quiet_river_4821

# Export to env vars for scripting.
$ darkcat identity export protonmail-quiet_river_4821 --format env

DARKCAT_ID_NAME='protonmail-quiet_river_4821'
DARKCAT_ID_PROVIDER='protonmail'
DARKCAT_ID_STATUS='confirmed'
DARKCAT_ID_HANDLE='quiet_river_4821'
DARKCAT_ID_PASSWORD='*****' # masked unless --reveal
```

The same set of actions is available from the TUI and the GUI's Identity dialog so you don't have to memorise CLI flags for routine vault chores:

```
TUI Identity screen (press `i`):
[L]aunch      – open the persona's signup URL through Tor again
[S]how        – prompt "Reveal?" and dump the persona record
L[i]nk        – pick parent + child personas
[U]nlink      – break a previously recorded edge
[E]dit        – credentials editor (see §21.5)
[N]ew, [C]onfirm, [B]urn, [R]efresh, [Esc]ape – unchanged
```

```
GUI Identities dialog (menu Identity → Open vault...):
New, Launch signup, Show, Confirm, Edit, Link, Unlink, Burn –
same actions wired through `invoke_cli_capturing`. The Show
button opens a Reveal / Masked picker before dumping secrets so
revealed credentials never land on the screen by accident.
```

Encrypted vaults (`personas encrypt`) are now first-class in the TUI and GUI: the first action against a `.pgp` vault opens a passphrase prompt, the verified value is cached for the lifetime of the dialog, and every follow-up action threads it to `cmd\_identity` via the same `DARKCAT\_VAULT\_PASSPHRASE` env var the CLI itself honours.

21.5 Editing credentials after manual signup

`identity new` generates a candidate handle / display name / locale; the provider's signup form may assign different final values. Use `identity edit` to record the real ones, plus anything the form only shows once (recovery codes, recovery email, BIP-39 phrase). The same surface is available from the TUI (`E` on the selected row) and the GUI (`Edit` button in the Identities dialog).

```
# Set the final handle the homeserver gave you, plus a recovery email
# already in the vault:
$ darkcat identity edit matrix-ashen_hawk_3760 \
    --handle "@alice:tchncs.de" \
    --email "alice@tchncs.de" \
    --recovery-email "recover@proton.me"

# Append two recovery codes (run again to append more):
$ darkcat identity edit gmail-quiet_owl_3221 \
    --recovery-code "aaa-bbb" \
    --recovery-code "ccc-ddd"

# Replace the whole recovery_codes list (Gmail re-issues every code):
$ darkcat identity edit gmail-quiet_owl_3221 \
    --recovery-codes-replace \
    --recovery-code "new-1" --recovery-code "new-2"

# Pass an empty string to *clear* an optional field:
$ darkcat identity edit matrix-ashen_hawk_3760 --notes ""
```

`identity edit NAME` with no field flags is a hard error so an empty TUI submit can't silently no-op.

21.6 Mail-provider presets for `personas add`

`mail send` / `mail check` need an SMTP/IMAP coordinate. Five presets ship the common ones so you don't have to remember the host / port / TLS combination – pick one with `--mail-provider`:

```
$ darkcat personas mail-providers
```

slug	site	notes
proton-bridge	127.0.0.1:1025	local Proton Bridge
disroot	disroot.org:587	STARTTLS / IMAP 993 SSL
mailfence	smtp.mailfence.com:465	SMTPS / IMAP 993 SSL
gmail-app-pw	smtp.gmail.com:465	requires app password
fastmail	smtp.fastmail.com:465	SMTPS / IMAP 993 SSL

```
# Persona for a ProtonMail account routed through the Bridge:
$ darkcat personas add proton-acct \
    --mail-provider proton-bridge \
    --handle alice@proton.me \
    --password "<bridge-app-pw>"
```

Explicit CLI flags always win over preset defaults – passing `--site` or `--notes` overrides the preset's value. Tutanota is intentionally absent: it has no SMTP / IMAP surface for third-party clients.

The same preset picker is exposed in the TUI and GUI so you don't have to drop to the shell for a one-off mail persona:

```
TUI:  press `p` (Persona) from the main screen, fill name + mail
      provider preset + handle, then Ctrl+Enter to add.
GUI:  menu Mail → Add mail persona..., pick a preset from the dropdown,
      click Add.
```

Both forms route through the same `personas add` handler, so the override precedence is identical to the CLI: explicit Site / Notes / Network fields beat the preset.

=====

Health & discovery

```
darkcat status
darkcat doctor
darkcat list all
darkcat list tor
darkcat seeds tor
darkcat discover "hacking forum"
```

Crawl & fetch

```
darkcat fetch <url>
darkcat crawl --seeds <url> [<url>...] --max-depth 2 --max-pages 200 \
    --topics kw1 kw2 kw3
darkcat crawl --seed-file /tmp/seeds.txt --max-depth 2 --max-pages 200
darkcat schedule add jobname --interval 21600 --seeds <url> [<url>...] \
    --max-depth 2 --max-pages 200
```

Search & analyse

```
darkcat search "phrase OR another"
darkcat search "wallet drainer" --strict
darkcat top --limit 25
darkcat top --protocol i2p
darkcat stats
darkcat history --url <url>
darkcat diff --url <url>
darkcat clusters --min 3 --limit 20
darkcat keys list
darkcat ocr <url>
```

Watchlist & alerts

```
darkcat watch add --sample 0day --sink log
darkcat watch add --target tor.taxi --sink notify
darkcat watch list
darkcat watch test <id>
darkcat alerts --limit 100
```

Liveness

```
darkcat liveness probe --known
darkcat liveness loop --known --interval 1800
```

Tor control

```
darkcat tor newnym
darkcat tor circuits
darkcat tor descriptor <onion>
darkcat tor bridges-add "obfs4 ip:port cert=..."
```

Identity Generator

```
darkcat identity providers
darkcat identity providers --slug protonmail
darkcat identity new --provider <slug> --transport tor --purpose "..."
darkcat identity new --provider matrix --instance tchncs # multi-host
darkcat identity launch <name>
darkcat identity confirm <name>
darkcat identity edit <name> --handle "@x:y" --email x@y --recovery-code ...
darkcat identity burn <name> --note "..."
darkcat identity list
darkcat identity show <name> [--reveal]
darkcat identity rotate-password <name> --print
darkcat identity link <parent> <child>
darkcat identity export <name> --format env
```

Personas (mail presets)

```
darkcat personas mail-providers
darkcat personas add <name> --mail-provider proton-bridge \
    --handle alice@proton.me
```

Chat

```
darkcat chat backends
darkcat chat login --persona <name> --network <net>
darkcat chat list --persona <name>
darkcat chat read --persona <name> --limit 50 -- <channel_id>
darkcat chat ingest --persona <name> --limit 500 -- <channel_id>
```

Export

```
darkcat export --format jsonl --since 7d
darkcat export --format stix --kind findings
darkcat export --format misp --kind findings
```

```
Frontends
darkcat          # CLI
darkcat shell    # interactive REPL
darkcat tui      # Textual full-screen TUI
darkcat gui      # Tkinter desktop window
```

23. GUI WALKTHROUGH – WHERE TO CLICK

Launch:

```
$ darkcat gui
```

23.1 Window map (where things live)

darkcatv0.4.0(banner – logo + tagline)

FileViewIdentityHelp(MENU BAR)

● tor ● i2p ● ipfs ● freenet ● gemini ... [RESCAN]

console

> topics []

> proto [▼ tor] pages [50] depth [2] threshold [0]

> actions [► CRAWL] [× STOP] > search [] [SEARCH]

> fetch [] [FETCH]

log

[crawl] fetched 35/200 ...

[crawl] depth=2 host=tor.taxi

[done] 200 pages, 18 findings

results

scorecategoryprotourl

0.91leaktor...

0.84markettor...

...

Ctrl+E export · F5 refresh ... [REFRESH] [EXPORT] [× QUIT]

← STATUS PILLS

← row 1

← row 2

← row 3

← left-click header to sort. Right-click a row for actions.

← bottom bar

Menu contents (exact labels):

File ▼	View ▼	Identity ▼	Help ▼
Export results... Ctrl+E	Refresh results F5	Open vault... ↑^I	Keyboard shortcuts...
Quit F1	Re-probe transports ^R		Run doctor...
	Database statistics ^I		About darkcat...
	Clear log ^L		

23.2 Day-zero health check

Click: View → Re-probe transports (or press Ctrl+R)
What happens: the dots next to `tor / i2p / ipfs / freenet / ...` flip green if the transport is reachable, red if not. The bottom message bar shows `► scanning...` while it runs.
If you want a verbose report: Help → Run doctor...

23.3 Crawl a built-in protocol pack

- > proto click the drop-down, pick `tor` (or `i2p`, `ipfs`, `gemini`, ..., or `all`).
- > topics type space-separated keywords, e.g.:
exploit cve 0day phishing
- pages change `50` → `200`
depth leave `2`

- threshold leave `0`
- Click [► CRAWL] (or press Enter in the topics field).
 - Watch the log pane on the left fill with `[crawl] fetched N/M ...`. To stop: click [× STOP] or press Ctrl+C.
 - When you see `[done]`, the right pane (results) is sorted by score, highest first. Click any column header to re-sort.

Note: the GUI crawls the built-in seed list for the selected protocol.
To crawl a custom seed list, use the CLI:
darkcat crawl --seed-file /tmp/seeds.txt --max-depth 2 ...

23.4 Fetch one URL (single-shot)

- In the > fetch row, click the text box and type the URL, e.g.:
https://tor.taxi/
or ipns://docs.ipfs.tech
or gemini://geminiprotocol.net/
- Click [FETCH] (or press Enter).
- The page is saved to ~/.darkcat/crawl.db; the log shows
`[fetch] 200 OK 12345 bytes`.

23.5 Search what you have crawled

- In the > search row, click the text box and type a query:
monero AND mixer
- Click [SEARCH] (or press Enter).
- Hits replace the right-pane table.
- Right-click any row to get a context menu:
 - Copy URL
 - Fetch this URL
 - Show mirrors (SimHash)
 - Show history
- Double-click a row to copy its URL to the clipboard.

23.6 Database statistics

Click: View → Database statistics (or press Ctrl+I)
A modal opens with: total pages, findings, per-protocol counts, top topics, watchlist hit count.

23.7 Identity Generator – create a ProtonMail account, click by click

This is the GUI equivalent of CLI section 21.

- STEP 1. Open the vault.
Click: Identity (menu bar) → Open vault..
Shortcut: Ctrl+Shift+I
→ A modal titled "darkcat – identities" appears.
It looks like:

darkcat – identities				
name	provider	status	transport	purpose
(empty on first run)				
[New] [Launch signup] [Confirm] [Edit] [Burn]				
[Refresh] [Close]				

- STEP 2. Click [New] (the leftmost button in the bottom row).
→ A small dialog "darkcat – new identity" appears with four fields:

darkcat – new identity	
Provider	
[▼ discord – Discord (social)]	
Instance (host)	
[▼ (provider default / N/A)]	
Transport	
[▼ tor]	
Purpose tag (optional)	
[_____]	
[Create]	[Cancel]

The Instance picker only lights up for multi-host providers (matrix, xmpp, mastodon). For everything else it stays on "(provider default / N/A)" and is ignored. Picking `tchncs` – German community homeserver – from the Instance list for `matrix` is equivalent to passing `--instance tchncs` on the CLI: the persona's site / network_or_domain become `matrix.tchncs.de` instead of the matrix.org default.

- STEP 3. Pick the provider.
Click the Provider drop-down. Scroll to:
 protonmail – Proton Mail (email)
Click it.
- STEP 4. Pick the transport.
Leave the Transport drop-down on `tor` (default).
(Switch to `i2p` or `proxy` only if you know why.)
- STEP 5. (Optional) Type a purpose tag.
Click in the "Purpose tag" box and type, e.g.:
 research-forum-X
This tag shows up in the table and helps you remember why the identity exists.
- STEP 6. Click [Create].
→ The dialog closes.
→ A new row appears in the identities table, e.g.:
 protonmail-quiet_river_4821 | protonmail | pending | tor | research-forum-X
→ The row is auto-selected (highlighted).
If something failed, a red error dialog pops up – read it, don't dismiss it blindly.
- STEP 7. Open the signup page in a Tor-routed browser.
With the new row still selected, click [Launch signup].
→ Tor Browser opens (if installed). Otherwise your default browser opens with HTTPS_PROXY=socks5h://127.0.0.1:9050.
→ The log panel behind the dialog prints the helper block:
 handle: quiet_river_4821
 password: 8x2-fJW9_zT...vGqM (copy now – shown once)
 display name: Mira Holt
 bio: security & privacy • EU-based, mostly evenings – no DMs
 signup URL: https://account.proton.me/signup
 no-phone path: protonmailrmez3lotcc...onion → email-only verify
- STEP 8. Complete the signup form in the browser.
 • Username → paste handle from STEP 7
 • Password → paste password from STEP 7
 • Recovery → leave blank or use a second darkcat identity
 • If asked for a phone: switch to the .onion (STEP 7 printed it) and select email-only verification.
Solve any captcha. Click "Create account" on Proton's page.
- STEP 9. Tell darkcat the signup is done.
Switch back to the darkcat window. The identity row should still be selected. Click [Confirm].
→ Status flips: pending → confirmed.
→ The log shows `~ identity protonmail-quiet\_river\_4821 confirmed`.
- STEP 10. (Optional) Edit credentials after signup.
Select the row → click [Edit].
→ A "darkcat – edit <name>" dialog opens with the persona's current handle, email, recovery email, display name, recovery phrase, recovery-codes count, and notes prefilled. Type into any field; leaving it unchanged keeps the current value, an empty string clears it.
 • Save – apply the changes (only the diffs are forwarded).
 • Rotate password – generate a fresh password and overwrite the old one. The new value is retrievable via
 `darkcat identity show <name> --reveal`.
 • Cancel – discard.
Recovery codes are appended (comma-separated). For Gmail-style providers that re-issue every code, edit from the CLI

with `--recovery-codes-replace`.

The TUI exposes the same editor: press `E` on the highlighted row inside `darkcat tui` → menu → Open vault.

- STEP 11. Burn the identity when the project ends.
Select the row → click [Burn] → type a one-line note in the prompt → OK.
→ Status flips to `burned`. The row stays in the table but no longer counts toward the per-provider cap.

23.8 Export results to a text file

Click: File → Export results... (or press Ctrl+E)
Pick a path in the file dialog. The current results-table contents are written as a flat text file.

23.9 Keyboard cheatsheet (Help → Keyboard shortcuts... or F2)

Crawl & search

Enter (in topics)	start crawl
Enter (in search)	run search
Enter (in fetch)	fetch URL
Ctrl+C	stop crawl

Identity / data

Ctrl+Shift+I	Identity vault
Ctrl+E	Export results
Ctrl+I	Database stats
F5	Refresh results
Ctrl+R	Re-probe transports

Help & info

F1	About darkcat
F2	This shortcuts dialog
Ctrl+L	Clear log panel
Ctrl+Q	Quit darkcat

=====
End of instructions.txt – see also docs/QUICKSTART.md, docs/USERGUIDE.md,
docs/NETWORKS.md, docs/INTERNALS.md, docs/CONFIG.md, docs/identity/.
=====